

Applied Category Theory

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okay so I guess we're there so let's get started so uh well gosh what an influence John John had in my life uh there I was a fledgling philosopher trying to uh interested in in mathematics and and he'd gone through his writings in the in the Long Way a long time ago in the 1990s gave me the confidence to think that you you could get access to the Forefront of of mathematical and phys scientific knowledge through his writings I mean what gosh what a what a joy that was to to engage with his writings back at that time huge influence on me and you know a bunch of people will know that we started a Blog together a while ago we had a lot of conversations over over a great many years so it's great pleasure to have him speak to us here today on on the play category Theory so thank you John well thanks very much thanks for inviting me here and uh the influence was reciprocal uh it's really exciting to think that what one is doing in mathematics could actually be interesting to a philosopher so that uh one isn't just like fiddling around with obscure equations uh so I want to talk about applied category Theory which is the latest turn in this game uh and I'll talk more about the history of the subject and what people are trying to do then uh any mathematics per se I've given lots of talks about the mathematics and so if all the people who I'll be uh referring to here so it's certainly easy to click on any of the links in my slides and get into the actual mathematics but I'll try to not say as much about that as I sometimes do so we often have this picture of different uh disciplines or subject areas talking to their neighbors and information flowing between pure math which is at some very abstract and uh to say the business World which is maybe at the very concrete end in stages with each discipline talking mainly to its neighbors so pure mathematicians talk to applied mathematicians they give suggestions to applied mathematicians and they find out about problems from applied mathematicians and learn techniques from applied mathematicians and similarly for each link down the chain with each uh discipline talking to its neighbors and in this model which of course is very oversimplified and and you can easily find a lot of faults in it uh in this model we could think of category Theory as the purest of pure some super pure mathematics that only can talk to

mathematicians who then relay their insights down the chain to the other more practical subjects um I think that's what a lot of people implicitly feel about category Theory or at least used to uh and so when people started talking about applied category Theory people laughed because it sounds like an oxymoron it just goes completely against this picture here however I think this picture is inaccurate for one thing a very important fact is that starting maybe around 1980 or so it became clear that computer science really cuts across this traditional flow of information computer science interacts almost directly with every single piece of this chain computer science has a big relation to category Theory which I'll explain but it also has a huge relation to other aspects of pure mathematics and Applied Mathematics and so on down to business so businesses will need to keep abreast of developments in computer science and so on basically we now communicate and think with the help of these machines called computers that we have devised to help us and and they changed the whole game at every single step of the way now I have this crazy idea that maybe category Theory actually really should be thought of in a somewhat similar way as interacting with all the different links in the chain from the pure end to the applied end rather than being tucked away at the super pure end of the spectrum now at first it seems ridiculous and are laughable I mean for starters category theory is sort of nothing compared to computer science everybody's walking around carrying cell phones nobody has even heard of categories by comparison so category two is an ant compared to the elephant of computer science however I'm thinking about what could be not what yet is and secondly when you think about how computers have unified the whole chain it's done so by means of the internet which is an enormous Network and category Theory isn't nothing if not the mathematics of networks or the science of networks it's other things too I shouldn't have said it quite that way I mean it's it is the science of networks but it does many other things too uh so so to really devise good methods of getting computers to do what we want you need a way to think about them and I think category theory is one of the important ways to think about them so you may think this is ridiculous still because for example if category theory was talking directly to business you'd expect that people would be going into business during category Theory or something like that setting up category Theory businesses that sounds sort of silly but in fact it's true so in fact people are setting up category Theory businesses and that's part of the kind of thing I want to talk about I want to talk about how category theory is now being used at all uh steps along the Spectrum and for that reason my talk will be less focused on the math than on the history of how category Theory became applied giving you a flavor of how it's being applied and pointing out some of the challenges

that show up um uh before I dive in I should say that this history is going to be extremely self-centered I'm not really in a position to write a good or talk about a good history of category Theory so hopefully someone who eventually tries to write a history of Applied category Theory we'll listen to this talk and fold it in as one data point uh I'll talk about a lot of people but I will also leave out a lot of people and I really apologize to all the people whose work on applied category Theory I'm not mentioning there's just no way for me to do a fair treatment of this big subject okay so I'll start arbitrarily in 1963 when Bill of beer introduced a concept called functorial semantics so it's a very general concept where you have a category that describes uh syntax describes ways of writing things down and then describe the process of interpreting those expressions or giving a meaning to those expressions the functor so from the functor will send expressions to their meanings but a crucial aspect of this whole setup is that there's not one god-given functor F that does this you can choose various functors from the category C to various other categories key uh because they're different ways to assign meanings to expressions and that flexibility is very important and it became important in computer so for example there's a nice there's a way to think about it where exceptionally well behaved programming language may give a category where you think of the objects as types or data types we've already heard about that idea so types x y z Etc and then morphism between objects like morphism f from X to Y we think of it as a program that accepts data of Type X is input and outputs data of type y and then the fact that you can compose morphisms uh tells you that you can hook up two programs to get a bigger program or the output of the first program becomes the input of the second and like that that combination that is a program in itself so I could say that like the relationship between the Lambda calculus which is sort of an idealized programming language simplified programming language and Cartesian closed categories uh understood by Lambic is an example of this Paradigm of using functorial semantics for uh for computer science but here I'm only so far talked about the category C I haven't talked about the uh the functor so what the functor could do well we could for example take a functor from our category to the category of sets not each data type X to a set and that would be the set of possible values of the of that data so like you could have a data type like real numbers and then definitely map it to the actual set of real numbers uh and then each program gets mapped ideally to a function and the key factor of a functor is that it preserves composition so that composing programs would compose the would give rise to composing the functions they compute and the identity program which does nothing at all you interpreted via this functor F it gives you the identity function which does nothing at all on a set so that's a

that's a paradigm that became important in theoretical computer science I wouldn't say that most uh computer scientists know or care about functorial semantics but in a certain community of the more mathematical uh theoretical computer scientists this idea has become quite important and there are certain languages like Haskell for example and others that that really take advantage of this Viewpoint in particular Haskell uses a category theoretic construct called monads which has sent many uh generations of young computer scientists running for help from a category theorists to try to figure out what a monad is I don't want to explain what a monad is it's a category theoretic construct which like many other constructs become important when you start thinking about programming in this functorial semantics way but in the 1980s something very different happened uh particle physicists realized that any Quantum field Theory specifies a category and now the objects are roughly speaking collections of particles and the morphisms are roughly speaking ways for particles to interact and turn into other particles and in fact Feynman drew pictures called Feynman diagrams for describing how particles interact with each other this was a bold move because although it seems very intuitive here you have a proton and an anti-proton coming in from the left and then they collide and all sorts of other particles shoot out uh in reality particles are not little dots moving along little straight lines quantum mechanics says that particles are wave like in character so when Feynman introduced this pictorial language he got a lot of opposition from older Quantum physicists like Heisenberg and Bohr and yet it turned out Feynman was correct in doing this what finally diagrams actually are later became clear much later is there pictures of morphisms in specific categories and each individual Quantum field Theory gives a category and then that determines what allowed find the diagrams you can write so if you have a theory that doesn't have quarks in it you're not allowed to write this q line for quark and then what do you do with these pictures Well your goal in particle physics is to compute a bunch of numbers like what's the probability that if I smack two particles into each other that some other particle shoots out so for calculating numbers we need uh to interpret our assignment diagram that is in the functorial semantics attitude we need to functor from our category to some other category which assigns meaning to the finding diagrams in this category we could take to be the category of Hilbert spaces a widely loved mathematical structure in quantum mechanics and so then it will this function will turn any Feynman diagram into a morphism in the category of Hilbert spaces and that's called a linear operator between Hilbert spaces and so you can look at what physicists are doing with Feynman diagrams uh as basically using them as a syntax which they then apply the semantics to to then

grind out numbers now Feynman probably would have laughed at this whole business because he knew what he was doing he didn't need any category theorists to tell them what he was doing but by the 1980s particle physics had become elaborate enough and deeply connected with enough pure mathematics that this attitude of thinking of upon field theory in terms of functorial semantics became a really profitable attitude you could do things with category theory that particle physicists were struggling to do without category Theory and so it helped and this led to a whole line of work uh trying to relate category Theory to quantum mechanics and in 2004 Samson Abramsky and Bob Coecke wrote a very influential paper showing that a certain Quantum process that goes by the name quantum teleportation and also other processes could be understood using category Theory and using diagrams that resemble Feynman diagrams but here the edges in these diagrams which now I guess we're reading from top to bottom rather than across the edges are not describing Elementary particles but just states of an arbitrary Quantum system so it could be an electron or it could be some larger system and I don't want to try to explain this in detail but there are but these are pictures of morphisms in some category and they're equal to each other and these morphisms describe some process where one participant always is called Alice in cryptography uh communicates a message to another participant it's called Bob and in quantum teleportation you have the superficial appearance that you're communicating uh faster than light but in fact what's really going on is that you've laid the ground for the communication by some classical process that has already transferred some information and then the rest of the information uh is transferred on the basis of that in any event quantum teleportation was a sort of magical wonderful thing that you can do in quantum mechanics and this gave a kind of category theoretic account for how it's working and that was quite exciting to people because it took the quantum applications of category Theory out of the realm of elementary particle physics and into a new realm that was developing which is quantum information Theory or Quantum computation which has the potential for being a quite practical thing and indeed already a Quantum cryptography is a is it is used so on the basis of this and other work and also on the basis I should say over a huge amount of Charisma Bob Coecke and Samson Abramsky built a huge group of people at Oxford sometimes called the quantum group working on category Theory the foundations of quantum physics and Quantum computation one interesting thing about this enormous group here is that it's in the computer science department so not a mathematics Department not even a physics department but a computer science department here's Samson Abramsky here and here's Bob Coecke looking sort of disreputable over on the

edge that's completely intentional he's like that and here are many many people including I see Jules Hedges who has mentioned uh in the previous talk that you were example categorical cybernetics um so Circa and co-authors developed this diagrammatic method for explaining quantum physics and they eventually decided it would be very good to write a textbook on this and so here's one of the two textbooks that Bob kruk helped write on pictorial methods of understanding quantum physics and the one interesting thing about this book so first of all I recommend this book if you're interested in learning category Theory from a unorthodox and sort of easier approach than the conventional one because it's very pictorial uh but I mentioned it because part of what what you wound up doing is hiding the category theory in the sense that you don't run around saying category and functor and all sorts of other jargon like that a lot instead you draw pictures of processes and you explain these pictures in a Hands-On way and you explain how you can stick one picture onto the other which is composing morphisms but you don't say the word composing morphisms necessarily because you don't need to uh and this is one of a generation I should say a new generation of category Theory textbooks that are part of the applied category Theory movement so if a category theory is going to reach directly to many different communities at different stages of the knowledge chain that I drew at the top of the talk uh it means that you can't force people to first become mathematicians before they then start doing things with category Theory the traditional approach to category Theory treats category theories like some finishing school that you engage in after you've already learned abstract algebra and this and that and algebraic topology and so on that's just too onerous a path to category theory if you're trying to get a computer scientists much less business people to get some inkling of category Theory and it's not really necessary to take that elaborate route because category 30 is something so fundamental that it can be explained more directly I especially recommend Eugenia Chang's book The Joy of abstraction which is a really wonderful introduction to category theory for people who have not had much prior exposure to mathematics at all and it goes all the way up to the Oneida embedding theorem thank you so in 2021 Perka actually left Oxford University and he became the chief scientist of a company called Continuum which uses category theory for Quantum Computing and also natural language processing this is a picture of him on the Continuum web page as you can see he's trying to quite hard not to fall into the stereotypical academic image here he's a great guy by the way you shouldn't mind me teasing him like this uh I think though that it is important it's having people with personalities who are willing to break out of the stereotypical mold of a math professor is important in this whole category Theory application movement uh I should

say that there's a lot of interaction between Quantum Computing ideas and natural language processing ideas which I won't really have time to get into uh to some extent they were visible in the last talk but if you want a real emphasis on the natural language processing aspect The Works of uh Tai Denae Bradley are very good she's another newer explainer of category Theory it's a non-category theorists and she is also working uh in a company that uses Quantum methods and categorical methods for natural language processing but Ontario semantics is extremely General so we it's not just applicable to the topics that I've already mentioned so in lots of areas of Science and Engineering people use a syntax where you draw diagrams and then a synaptics where you convert those diagrams into math which is very often the systems of differential equations you're describing how things change in time you're describing dynamical systems uh so you use differential equations but the interesting part is that hooking up diagrams to each other is actually composing morphisms in a category and the fact that you're working with frontorial semantics says that when you compose those morphisms in the syntax category your functor will map it the resulting composite to the composite of the two systems of differential equations that each picture separately is mapped to so in other words there's another category lurking around here where the morphisms are actually systems of differential equations systems of differential equations with some specified inputs and outputs to surface as the source and target of your morphism here are some different pictures you've seen a bunch of these types of pictures I hope so electrical circuit diagrams are one of the earliest and very widely used this thing here is a called a petri net and it's a petri net of a particular model for the AIDS virus and its interaction with blood cells infection of white blood cells this here is a very small uh diagram in the subject of control theory where you have some process in an industrial plant that you're trying to keep under control and so you take feedback you read off information from that process and you use it to feedback affect the the process itself to keep it on track I mean the very simplest example here would be like a thermostat but in in industrial chemistry they become much more elaborate and so there's a whole diagrammatic language of control theory that you can read about in textbooks on that subject and this is a little diagram of some molecules in molecular biology molecular biology or biochemistry they have so many different diagram languages that in fact they've set up a committee on to standardize the diagram languages so that people reading different biochemistry papers can look at the diagrams and not have to be explained each time what the diagrams mean but it's sufficiently Baroque that they uh that the committee wound up choosing three different diagram languages because they they need multiple languages to

describe different things they're interested in so these are very widespread and the people in these different subjects don't think of what they're doing here as category Theory they don't think of it as mathematics they think of it as this is the stuff we draw to understand the differential equations that you can convert these pictures into the equations of the math but these pictures are our own private little trick for understanding what's going on but I claim that these pictures are actually also mathematics and they're these diagrams are morphisms in various categories which are using functorial synaptics to convert to differential like excited that had to be true and I wanted to work out the details around 2012 so I asked Brendan Fawn who eventually became my grad student to tackle what I thought would be a very good first example namely electrical circuits because everybody knows about electrical circuits and the mathematics of them is in substance very well understood linear electrical circuits for example made out of just resistors or maybe capacitors and inductors as well I asked them to create and study a category having open electrical circuits as morphism so these are electrical circuits which have certain things hanging out which serves as inputs and outputs I think in the last part they are called cilia so these are just these dots here um so it took forever to do it so it seemed but it turned out that what he was doing was not just answering the question that I asked but in fact uh answering a much more general question he invented a general theory which he called decorated Coast bands for handling this problem but also all the other problems that I described that is essentially all the other kinds of diagram languages all right showed can be understood using the theory of decorated co-spans although the details have to be individually worked out in each case this decorated coastband formalism is a is a recipe for getting categories of this General sort and so naturally it did take him a while to do a good job of it and he finished his thesis on it in 2016. I won't attempt to explain decorated post bands any more than I just have I've given lots of talks on decorating post bands and you can see on YouTube me giving talks about them if you're interested I just want to mention one little thing is that well first of all here's Brennan Fong and here's another student of mine Blake Pollard who together with Brendan applied this theory of decorated close bands to the second example that we tackled which was Markov processes did a lot of work on Markov processes uh and another thing I want to mention is that uh Brendan went to get his PhD at Oxford so he was in the enormous group that I mentioned and I believe Kirko is his de facto it was his day jury advisor but in fact I was the the de facto advisor telling you important to Brendan about this this problem uh and so nonetheless it's really important that this large body of people that uh at uh at Oxford is is spreading across the world doing many things and Brendan is one of them so

after doing this doing his PhD uh Brendan went to MIT in 2016 to work as a postdoc with David Spiveak so Spivak is another very important figure in applied category Theory so among other things he had helped to create a language for databases called functorial query language which has now become the basis of a company called Connexus AI with Ryan Wisniewsky as a as a head of that company but David Spivak also involved this approach the databases really relies heavily on functorial semantics so the idea is the database schema is is an object in some category and then by a functor you can map it to a sort of populated or inhabited database schema where where the the your table of is actually filled with entries you can imagine like taking a spreadsheet and actually typing in the entries that's one possible uh interpretation of of the abstract database that has not yet been interpreted with with specific people's names and numbers and in 2014 David wrote a textbook on this idea and others called category theory for the Sciences so I also highly recommend this as yet another uh of this new generation of category Theory textbooks let's try to reach out beyond the traditional confines of the category Theory community of Highly pure mathematicians toward other audiences and in 2010 2018 uh Brendan Fong and David Spivak wrote another textbook explicitly on applied category Theory which is uh fondly called seven sketches but its actual title is here earns larger uh emphasizing the principle of compositionality in many guises and they taught a course on it which you can see on YouTube if you click on this link in my slides and the textbook is also available for free on the archive as well as for purchase and I'm mentioning this stuff about things being available online for free partially because when you're trying to reach out of the pure math realm into the rest of the world and teach the rest of the world category Theory you quickly realize that making things easily available for free is going to speed up the process a lot and so I think there's a general commitment so far on the part of a lot of Applied category theorists to make things open source and uh easy to get at now in 2018 my graduate student Kenny Corser there and I came up with structured Coats Fans which is a simpler alternative to decorated Coats bands turns out to be not quite as flexible as decorated Coats Fans but in many cases either formalism works and in those cases I'd say structured co-spans are simpler in 2020 Christina Vasilakopulu who's standing to the left there uh helped us understand how decorated and structured Coats bands are related for a while I thought structured Coats Fans would replace decorated Coats Fans but it turns out we we now see that they can't cover quite as much territory and that we really need decorated co-spans in some case cases so the full story is somewhat elaborate right now uh I've given talks about that but it's it's actually very pretty so by now structured and decorated Coats Fans have been used to study open systems of

many kinds so by open systems I mean systems that have loose ends that are meaning systems that are amorphisms in a category that you can stick together or compose to get bigger systems and where open graphs are just a simply a graph with Loose Ends it's very combinatorial structure that's a very simple example open Petri Nets is a more useful example here's a picture of a an open Petri net where you have uh things describing up they're called called places that could represent like populations of different kinds of chemicals or organisms and then you have these blue boxes called transitions which describe how things can come out of various places and turn into things of other kinds um if you attach numbers to these transitions uh describing the rate at which the transition tends to occur you get something called a open Petri net with rates or maybe an open chemical reaction network with rates uh and my student Blake Pollard worked on those open electrical circuits I've mentioned open Markov processes I mentioned open dynamical systems those at the systems of differential equations interpreted as morphisms in some so there are papers on all of these and if you click on these Brown things you'll get to those papers so at a certain point I thought that maybe someday all of this would lead to some unified language of the Sciences and my ambition which seems very limited in retrospect was that by interacting with Scientists maybe eventually this unified language would help scientists in one discipline talk to scientists in another discipline and share tricks of various sorts but then something quite different happened which is much more exciting in my mind namely in 2019 James Fairbanks is an engineer and Evan Patterson who had statistics PhD began developing a computer system called algebraic Julia for it's a system for high performance scientific computing built on the language called Julia which is designed for crunching numbers really fast but algebraic Julia lets you program it using constructs from category Theory so in algebraic Julia you can talk about categories functors and all these other things that category theorists like to talk about and write your programs using those contents so it's a very very high level system but coupled to the raw computational power of algebraic Julia so now we're attractive revolutionize scientific Computing in this way and in 2020 Evan implemented structured co-spans in algebraic Julia and then using this system uh for the example of open Tech units he and Mika halter reconstructed some of a covid-19 model which was currently being used by the UK government so the UK government had a very elaborate model of covid-19 with many populations of different people with different conditions possibly living in different locations Etc uh and then I guess sorry the populations of these orange boxes and able to turn into other kinds of people via these transitions in blue the whole model is much larger than this this is just a tiny fraction of it and what they wanted to

demonstrate was that you could build the whole model by building smaller models and pieces and then subsequently sticking together those pieces which is in fact something that had not been done before models were sort of had to be constructed in whole cloth by essentially by one person keeping track of the whole thing but this is a new approach compositional epidemiology and so it attracted the attention of some epidemiologists especially since covid-19 is a very live issue at the time then in a different direction in 2021 Brendan Fong and David Spivak who had been remembered over at MIT decided to quit traditional Academia and start their own Institute called the topless Institute in downtown Berkeley it's an Institute for Applied Mathematics but with a heavy slant on applied category Theory and they hired a number of people including Evan Patterson and also Sophie libkin who's here who will appear later in the story and it's a real real incubator for our ideas on a flight category Theory now and so this is another one of the sociological moves that I had to be made I think to bring applied category Theory forward I recommended Brendan not try to start an Institute right after his first postdoc but luckily he was much braver than I would ever be and he went ahead and did it and it's quite successful now so in 2022 Sophie Lipkin Mika halter Evan Patterson James Fairbanks and another person I haven't mentioned Andrew boss used algebraic Julia to systematically build models of epidemia epidemiology using open Petri Nets with rates so here are some open Petri Nets the rates that is in the number data is not being shown here and here they're being stuck together to form a larger open Petri net the way that they're being stuck together you'll notice is not end to end this is not the normal end-to-end composition that you see in a category this is something a little bit more General we're sticking them together using a trick called an operand so operatic composition is one of the things that they Incorporated in their algebraic Jewelers software which is this was done this previous work was done in a slightly at hot basis but but this really used operats in the systematic way and then in 2022 also another team of people including myself used to algebraic Julia to create software for building models of epidemiology using another style of diagrams called open Stock flow diagrams well stock flow diagrams but we're making them open so here are some of those kind of diagrams being stuck together in an operatic style that is not end to end but sort of gluing them all together to form a bigger model so part of the point here is that when we started talking to actual epidemiologists we applied category theorists learned more about how they actually want to do their models and they didn't really want to do their models using pet units primarily they wanted to use a different technique that they considered congenial called stock flow diagrams um so in fact there's a community of epidemiologists who use

stock flow diagrams to model the spread of disease and it includes my co-authors Nathaniel Osgood and his grad student xiaoyen Lee and one crucial fact is that they did covid modeling for the government of Canada so they were requested by the government of Canada to set up covid models for a number of different provinces and in the process they gained a lot of experience uh with stock flow models they already had a lot of experience and they wouldn't have been asked to do it but they gained a lot more experience and they learned some of the problems of the existing technology this is a beautiful thing about stock flow diagrams which is that with a little bit of just a little bit of training you can read them and you can understand what they're talking about so a fellow named Hoffman has pushed the idea of community-based modeling where you go to a community that has some problem like a disease or some other type of problem and you help model it and you model it by drawing stock flow diagrams which here are hanging on the wall in a room in rural India and that people can look at those diagrams and comment on them and help you improve the model because these people the local people they will have the knowledge of what's actually going on that the modeler who just flew in from Oxford or wherever may not have and here we see a woman here looking at the picture on the wall and this is actually from a story where this woman noticed that some ingredient was being left out of the model uh that these Village Elders hadn't known about or had thought about and so she helped fix the model so that's the example of what you can do and you'll notice that none of the people in this story here had to know category Theory but they were in fact uh drawing morphisms in the category on the notes to themselves now most stock flow modeling in the commercial level or the hype or the government level is done using a piece of software called any logic which is very powerful but it has several big problems it doesn't let you compose models it lets you make a model but it doesn't let you take a bunch of models and stick them together it doesn't separate syntax from semantics it only has one semantics namely the differential equation semantics which is the most important semantics but there are others that you would like it has no support for what's called stratifying models that term was very confusing to me it's sort of charging in on this community what it means is taking a model and splitting One stock one type of group of people into several stocks so for example you might have a simple model and then you may say oh we should subdivide people according to age groups because older people who get the disease more more than younger people or something like that and you would like your software to make it easy to do that um it had no support for collaboratively building models it's basically one person as the software and they make the model they could talk to other

people but they're the one who's building the model and it's not free it's commercial piece of software and it's not open source so people can't get in there and modify it and improve it our new work and to fix all these problems and I could say we're well on the way to fixing all these problems so the reason it's so important to produce models is that the models that get actually used in the real world are immensely complex so here's a picture of Osgood and Lee's covid model this is a page in any logic the software this is the whole model as you can see it's an enormous web of of stocks and flows and such and you you really would prefer actually not to have to deal with this whole entity at at once you prefer to have it be thought of as being made up out of smaller pieces because mentally it is made up out of smaller pieces uh but but that's not captured by the the software unfortunately any logic just doesn't do that for you so we've not created software that does that it supports compositional modeling with the stock flow diagrams so it's called stock flow and if you click on this link you'll go to GitHub which is where computer programmers keep their open source software so you can get download the software and so it lets you create and compose open Stock flow diagrams and lets you stratify them that is subdivide stocks into smaller stocks for those of you who like category Theory I'll mention that that's done using a technique called pullbacks in category Theory it lets you choose among several different semantics functors so one of which is the differential equation semantics uh and of course it lets you solve the differential equations once you get them and then Nate Osgood and another student of his Eric redika went further and made a graphical user interface for stock flow so stock flow is written in this programming language called algebraic Julia to interact with it you needed to know algebraic Julia but this graphical inner graphical user interface called Model collab it runs on a browser and you manipulate the the model by actually dragging and dropping little boxes and drawing little edges between boxes still uh and because it runs on multiple browsers teams of different people located in different locations can a lot can collaborate to build stock flow diagrams so I could build my piece you could build your piece we could both save them they'd be saved on the on the cloud and I see there's this little thing called Cloud here uh that that's not what it is actually for them sorry that's a mistake uh but but you can you can say it's gonna laugh at that um actually clouds are these little funny things here uh but you can say you can save your models on the uh on the internet and so that other people in different locations can access them and then glue them together to form larger models so in the goal to really apply mathematics there are many levels because the next step is to actually train epidemiologists to use model collab so we have some things going for us so first of all eight Osgood runs regular training sessions which goes by the

Quasi militaristic terminology favored in computer science of boot camps because they're supposed to be intense so it runs boot camps on epidemiological modeling so he is beginning to teach people model cloud second no understanding of category Theory or the programming language Julia is required to use model collab so it's really crucial when you're applying category three that you can ultimately Black Box the category theory in it so that people do not explicitly need to know it to use what you're creating although they may be using it in some kind of implicit way uh and third it's free while any logic you have to pay for which should ultimately be an advantage but I should still say it will take a lot of work because when you have a generation of modelers trying to use any logic this new paradigm is not so easy to shape so as I just already said applying mathematics takes many steps and as I've moved from being more of a pure mathematician towards a bit more at least some some days of the week of an applied mathematician I realize how easy peer mathematicians have it you just write a paper you explain your idea you throw it out there and you hope that somebody reads it but when you're trying to actually carry out the full application of an idea you have to interact with many people with many skills persuade many different communities that what you're doing is actually worth doing and it's much more elaborate but I should also say very satisfying endeavor laughs so I'll conclude by saying that applied category theory is interesting in many ways some of which I hope will be interesting to philosophers uh so some of these ways are separate from sort of the core mathematical ideas so one thing as I hope I've illustrated here by my somewhat self-centered history is that it requires new forms of communication and collaboration because they're talking to new types of people and luckily uh these are starting to come into existence so there's an annual applied category Theory conference that's first started in 2018 uh run that time by Brendan Fong and some of his other young friends another is a journal so there's a journal called compositionality which is again started by Brandon falling in some of his collaborators and friends this is an open access free to publish free to read Journal of Applied category Theory which I recommend to your attention and another is called the category Theory Community server which is sort of like a chat site or a discussion site for Applied category Theory actually for all category three but it has a lot of Applied category theorists there uh this was initiated by my grad student Christian Williams and so these are different ways for category theorists to start talking to other kinds of people people who want to might want to apply category Theory or learn category three and I hope that these catalyze the the process done trying to outline here another thing is that category Theory raises ethical issues in ways that category theorists aren't used to pure mathematicians tend to feel

themselves somewhat insulated from a lot of ethical issues because what although their math make it used for all sorts of purposes they don't consider themselves particularly responsible for how their mathematics gets used because it's so abstracted they sort of can blame someone else for understanding it and doing something with it but when you start doing uh explicitly applied to category theory that sort of veil of uh protection is is ripped open and you see that what you're doing may have a fairly immediate effect so for example the biggest funder of research on applied category theory is currently the US military right now and also a lot of Applied category theory is being used for artificial intelligence or machine learning or large language models or so on so I'm not saying that those are necessarily bad things I'm saying that those are things that definitely raise serious ethical issues and so I'm worried about the danger that applied category Theory will wind up merely intensifying deal already existing patterns in society that is it will only make the richest and most powerful people even richer and more powerful I would prefer it if applied category Theory could do something really different really new this is one reason why I've tried to focus on epidemiology which is a little bit different than than some of the other uh applications of course even at the demiology ultimately feeds into the healthcare system so there are questions about that as well I would really like to use applied category Fury for environmental issues uh but I haven't quite gotten there yet anyway sorry I just wanted to emphasize that uh you know ethics is the provenance of philosophers and this is a really interesting intersection of mathematics and ethics which I hope that people think about and another thing about category three does is Revitalize what is a very old question which mathematical structures are best for describing and designing systems I emphasize designing because with the rise of computers whatever you can think about clearly enough you can make so it's not merely describing the external world it's you're also making worlds based on your descriptions and so this immense Freedom that you have in computer science that the question is what's what should guide you and it opens progress it opens possibilities for some really radical progress because these are some really brand new ideas that we're talking about here that may let us organize systems in new ways so I think there's a lot to think about here and I hope I hope some people here uh give it a try okay I think I'll declare it done brilliant you can see why I got drawn in by him 30 years ago it's uh his ways of conveying ideas so clearly okay um do let's have some some questions then let me bring up the people perhaps if people online have some questions they'd they can raise a hand and I perhaps we should be able to see or anyone here uh okay uh Mark's but I I do have a question and I'm I'm finding stuff that I've thanks for that by the way um this is really

exciting stuff for me but I'm wondering if I'm biased and reading too much into this that uh uh that just works in my self-interest um I do philosophy of time and um I'm one of the uh the few people who defend the notion that time flows and uh um I see in your work uh but I'm uh maybe it's through these biased eyes that uh this uh this project is bringing back the idea of uh of thinking of uh uh differential equations for instance as things that flow uh that functions or something that flow and I noticed the importance of diagrams as uh and these diagrams of uh uh uh are reminiscent of what organizations would call flow charts and uh um I'm I'm wondering whether I'm reading too much into this as something that might be an answer to the the static view of uh of uh in in in maths that came out of uh set theory and uh uh um thinking of uh um functions as uh as something that uh is static and not doesn't float so I'm I'm wondering whether or not I'm reading too much into it so maybe maybe that's the question sure um yeah the question of static versus Dynamic is as you know very well or extremely tricky question because I could take any dynamical system and when it's when it's done or when we imagine it as being laid out in time then we can think of it as a static thing with time is just another space like Dimension and and I think there's three it's very hard to like definitively break out of that uh so but nonetheless category theory is a generalization of set theory in a sense in which instead of just having elements in a set you have objects and also morphisms and the morphisms have are meant to convey some kind of dynamic that aspect of the system that is a amorphism goes from an object to an object and that's a very primitive sense of Dynamics there of the the start and the end and and and it brings a kind of dynamical quality to to the foundations of mathematics which is is a real thing it's not it's not just an illusion uh I mean even though you could if you ask a traditional mathematician what is a category and there's a oh it's a set of objects and a set of morphisms and some functions doing this and that right so you could say oh or just back in the same old Paradigm and that's what I meant by it's hard to definitively break out of that Paradigm I I think it's more in some ways more of a question of attitude rather than than anything else and anyway I think the interesting thing about the differential equations here may be the interesting new thing maybe the study of open differential equations which are differential equations where some quantities are explicitly labeled as affected by the outside world affected by the part of the world which is not modeled by the differential equation so uh traditional differential equations well no physics often likes to study what are called autonomous systems of differential equations where you know everything there is to know once you know the differential equations uh where as open differential equations which I should say are also studied by persistent

especially Engineers have certain quantities which are not determined by the differential equation they're regarded as being affected by the outside world and and that that on the one hand allows you to treat systems differential equations as morphisms that it could stick together by sticking some of these quantities on the edge to some other quantities on the edge but it also means that you're you're not just dealing with a deterministic system in the classic sense anymore because you have quantities that are affected by the unknown so so they it definitely changes your attitude towards differential thanks uh a question from Toby it's not so much a question sorry although I would love to I mean there's lots of Expressions I could ask but I need to head off because I actually have to go give this talk again just now thank you very much for inviting me and uh I hope the discussion is profitable okay thanks baby uh I had a um I was telling the interests a lot of people around here is is the difference between when you get variables that are just sort of associated with each other there's some kind of statistical Association versus when you think there's some kind of causal connection between them uh and you might think you might think you can go much better grip on the world if it isn't just Association data but there's a you've got a you know that a causes B other ways of getting at this through your um or is it is it that you you I guess the model is a kind of coming already with their sense of what causes wild and is there any way of representing the difference yeah so this particular line of work that I was outlining as you say I was mainly most of my talk was about model building where you you have your opinion about what causes what and how it affect what how X affects Y and you build the model into that so there's a flip side of the whole um modeling process which I didn't talk about at all which is incredibly important in epidemiology which is you know inference so you you measure data and you try to establish causality and also you know various numerical quantities which you would then put into your model um so so there's a lot to applied category theory has to offer to that and I'm afraid I'm not quite as uh knowledgeable about that I will say that um Brendan Fong for his master's thesis uh worked on a categorical approach to uh Bayesian networks and so and and and that is subsequently grown into things like what uh Toby was discussing uh and so that should that's that's also part of the big story and another part of it is that there's a technique called uh particle filtering where you which is widely used by epidemiologists where once you have one of your models of the sort that I've been discussing you run it and you run it a bunch of times and you see what kind of outcomes it predicts but then you use that to uh to compare to reality and you use that to estimate the parameters that you'd be in your model so so I guess what you could say is that like what I've been describing was just sort of one half of a feedback loop of

interaction between modeling and then comparing model with reality and and yeah so the whole thing should be should be carefully understood and I'm afraid I have sort of focused on one half of it sure yeah I've seen various diagrammatic techniques from category theorists and really get it kind of causal modeling I mean that exists out there doesn't it yeah it does and that's important okay um well there's I think there's no further questions when we reached six o'clock anyway so let's just oh sorry was there one there was a one sorry there's Dominos yes sorry I I I'm casting around I have lots of things to say sort of trying to formulate them into the questions typical hi John uh um so one thing that in um I know in computer science is quite a big deal is sort of algorithms and data structures um and kind of getting the right data structure for the right task to get good performance um and I think you know there's a lot of things that category theory has to say about this um uh kind of describing data structures in in a more abstract way sort of polynomial functions and things like what I haven't quite sort of understood is how much of that kind of work is going on in in the applied category sort of movement recently sort of linking with with back back with computer science the sort of like programming aspects around what I would call kind of classical computer science of algorithms and data structures um do you know if there's much of much of that going on within the sort of act World um or is that um largely more back with the compute scientists um so the algebraic Julia Community is really trying to blend computer programming with category theory in a really rich way so that you can write software and you can say you know Define functor define pre-shift category blood so that that and that's just one of several different initiatives to try to bring category Theory into programming so not just have it help programming it but I would actually be part of the programming so that that's one thing there are lots of other things and I probably don't know the other things so I will just mention again this company uh connexus.ai uses uh categorical approach to databases so it turns out that the updating databases is very nicely described using categorical ideas like on extensions and um Mike Johnson in Australia has been serving as a consultant for uh for database companies using category Theory ideas uh for many years but now this company based on David spivex work is is also doing that uh and there are and there are a lot of other things I should say that there's just a lot more on the computer science side that I'm not very familiar with okay all right okay thank you very much okay so well here we've reached home so let's all we have to do is thank John very much for his for his very interesting talk [Applause] so I've recorded the session and we can you said you're thinking putting it online we could yeah I'd like to if that's acceptable trip yeah yeah let's do that good all right okay well thanks thanks ever so much

John okay thanks very much bye bye everyone online bye I'll see you see you look like
[Music] all right